



ADBMS-Expt 2

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Experiment 02

Level: Easy

- **(A)**

A university administration requires an analytical performance report for each academic unit. Write a PostgreSQL solution to calculate the average marks scored by students grouped by their respective departments. The solution must join the student profile, course subjects, and grade marks tables, returning a breakdown sorted by the highest average marks down to the lowest.

Output

Department	Average Marks
Mathematics	95.00
Computer Science	83.33

Servers

Dashboard

Properties

SQL

Statistics

Dependencies

Dependents

Processes

Employee/postgres@PostgreSQL 18

Employee/postgres@PostgreSQL 18

No limit

Query

Query History

```
26 INSERT INTO Departments (dept_id, dept_name) VALUES
27 (1, 'Computer Science'),
28 (2, 'Mathematics');
29
30 INSERT INTO Students (student_id, student_name, dept_id) VALUES
31 (101, 'Alex', 1),
32 (102, 'Bella', 1),
33 (103, 'Charlie', 2);
34
35 INSERT INTO Subjects (subject_id, subject_name) VALUES
36 (501, 'Data Structures'),
37 (502, 'Calculus');
38
39 INSERT INTO Marks (student_id, subject_id, marks_scored) VALUES
40 (101, 501, 85),
41 (101, 502, 98),
42 (102, 501, 75),
43 (103, 502, 95);
44
45
46 select d.dept_name as Department, round(avg(m.marks_scored),2) as Average_Marks from Students s
47 join Marks m on s.student_id=m.student_id
48 join Departments d on d.dept_id=s.dept_id
49 group by d.dept_id
50 order by average_marks desc
```

Data Output

Messages

Notifications

Showing rows: 1 to 2

Page No: 1 of 1

	department character varying (50)	average_marks numeric
1	Mathematics	95.00
2	Computer Science	83.33

Total rows: 2

Query complete 00:00:00.081

LF

Ln 50, Col 28



Experiment 02

Level: Medium

Two legacy HR systems (A and B) have separate records of employee salaries. These records may overlap. Management wants to **merge these datasets** and identify **each unique employee** (by EmpID) along with their **lowest recorded salary** across both systems.

Objective

1. Combine two tables A and B.
2. Return each EmpID with their **lowest salary**, and the corresponding **Ename**.

Table A

EmpID	Ename	Salary
1	AA	1000
2	BB	300

Table B

EmpID	Ename	Salary
2	BB	400
3	CC	100

Servers

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X Employee/postgres@PostgreSQL 18 X

Employee/postgres@PostgreSQL 18

Query

Query History

```
14
15 INSERT INTO A (EmpID, Ename, Salary)
16 VALUES
17 (1, 'AA', 1000),
18 (2, 'BB', 300);
19
20 INSERT INTO B (EmpID, Ename, Salary)
21 VALUES
22 (2, 'BB', 400),
23 (3, 'CC', 100);
24
25
26 SELECT
27     EmpID,
28     Ename,
29     MIN(Salary) AS Salary
30 FROM
31 (
32     SELECT * FROM A
33     UNION ALL
34     SELECT * FROM B
35 ) t
36 GROUP BY EmpID, Ename
37 order by Ename;
38
```

Data Output

Messages

Notifications

Showing rows: 1 to 3

Page No: 1 of 1

	empid integer	ename character varying (50)	salary integer
1	1	AA	1000
2	2	BB	300
3	3	CC	100

Total rows: 3 Query complete 00:00:00.056 LF Ln 37, Col 16



Experiment 02

Level: Hard

A multinational organization maintains separate records for employee information and department details. The management wants to identify the employees who receive the highest salary within their respective departments for an annual performance report.

If more than one employee earns the same highest salary in a department, all such employees should be included in the report. The final output should display the **Department Name**, **Employee Name**, and **Salary**, arranged in ascending order of department name.

Note: The employee and department information are maintained in separate normalized tables.

Input Table: Employee

ID	NAME	SALARY	DEPT_ID
1	JOE	70000	1
2	JIM	90000	1
3	HENRY	80000	2
4	SAM	60000	2
4	MAX	90000	1

Department

ID	DEPT_NAME
1	IT
2	SALES



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Experiment 02

Part B: FIND THE 2ND HIGHEST EARNING EMPLOYEE from the same schema.

